Luchs B interfaces: VCU reference and draft LCI

Developer guide V0.1 | Source software v1.1.0 | 2026-10-11

Separate B VCU V2.6 and draft LCI software references. Confirm the delivered interface before connecting.

Status: Draft interface reference

Choose by the delivered interface, not the default

The software package contains two different B codecs. legacy_b targets the VCU upper interface V2.6. lci targets the LCI v1.0 draft design. The default B launch selects lci; a default is not evidence that a chassis has LCI firmware.

Property	VCU V2.6 / legacy_b	LCI v1.0 draft
Status	Unit-specific VCU reference	Firmware design / simulator reference
Byte order	Little-endian	Big-endian
Command level	Accelerator/brake and steering; host speed loop	Body-motion/axle requests; design closes loops in chassis
Bitrate	500 kbit/s expected, to confirm	Draft specifies 1 Mbit/s on customer CAN
Feedback	Coarse speed, gear, battery and fault categories	Design includes motion, live limits, IMU and extended state

Scope note. LCI heartbeat stops, maintenance recovery, IMU, payload switching and live derating are conditional design features. A simulator demonstration does not confirm them on delivered hardware. This guide publishes no new firmware or reviewed B spreadsheet.

Read the VCU reference with its open assumptions

Source reference	Meaning	Boundary
0x701 UpperCmd	Host accelerator/brake, steering and control request	The supplied sheet writes 701; confirm the numeric radix
0x703 VCU2Upper	Battery, gear, mode, coarse speed and fault category	Confirm the actual ID on the unit
Fault category 0 / 1 / 2 / 3	None / motor controller / BMS / several	Not LCI severity and not a detailed fault-bit table

The public driver source identifies steering sign, current offset, frame numbering and bitrate as items to confirm. Do not convert a software default into a verified vehicle specification.

VCU odometry uses speed in 1 km/h steps and the commanded steering angle. It is a coarse software estimate; do not treat it as precision localisation feedback.

Understand the LCI draft without deploying it

Draft ID	Design purpose
0x5A0 / 0x5A1	Body-motion and steering requests
0x5A5	Host heartbeat, alive counter, requests and checksum
0x5B0 / 0x5BD	Live limits and chassis state
0x5BE / 0x5BF	Raw IMU rates and version/model identification

The draft defines a 20 ms heartbeat and 200 ms timeout, with a counter and checksum. These are draft protocol requirements, not measured stopping performance or proof of an installed feature.

Do not merge a little-endian VCU field into the big-endian LCI decoder. Match the DBC, codec and firmware as a set. LCI fault levels describe severity; VCU categories identify broad fault sources.

A software-only comparison

Choose one command to explore the packaged simulator. Both keep dry_run:=true:

```
ros2 launch luchs_bringup luchs_b.launch.py \
  protocol:=lci dry_run:=true
# Or inspect the legacy interface simulator:
ros2 launch luchs_bringup luchs_b.launch.py \
  protocol:=legacy_b dry_run:=true
```

Record the software configuration and observe the reported state. For physical hardware, first obtain confirmed interface, harness, bitrate, identifier radix and steering-sign information for that unit.

Scope note. B geometry, steering range and loaded wheel radius remain unresolved in the public source. This guide offers a way to read that source, not a new approved datasheet, B engineering model or maintenance procedure.

Source and edition

Prepared and reorganised by RJ Tech from the already-public Luchs ROS 2 v1.1.0 package by TerraAgent Robotics. Source licence: Apache-2.0. Source text and examples were shortened, explained and scoped for this topic. Original archives and protocol files remain unchanged.

Source file: docs/protocol_luchs_b.md

Source file: dbc/luchs_b_vcu_v2.6.dbc

Source file: dbc/luchs_lci_v1.0_draft_r3.dbc

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